a

PRIMARY ESTIMAND (this paper)

frozen S0: PCA rotation + ridge
estimated ONCE on IMvigor210,
then applied transform-only to the
target cohort (no target labels,
no re-estimation)

different estimand →

LOCO ESTIMAND (sensitivity)

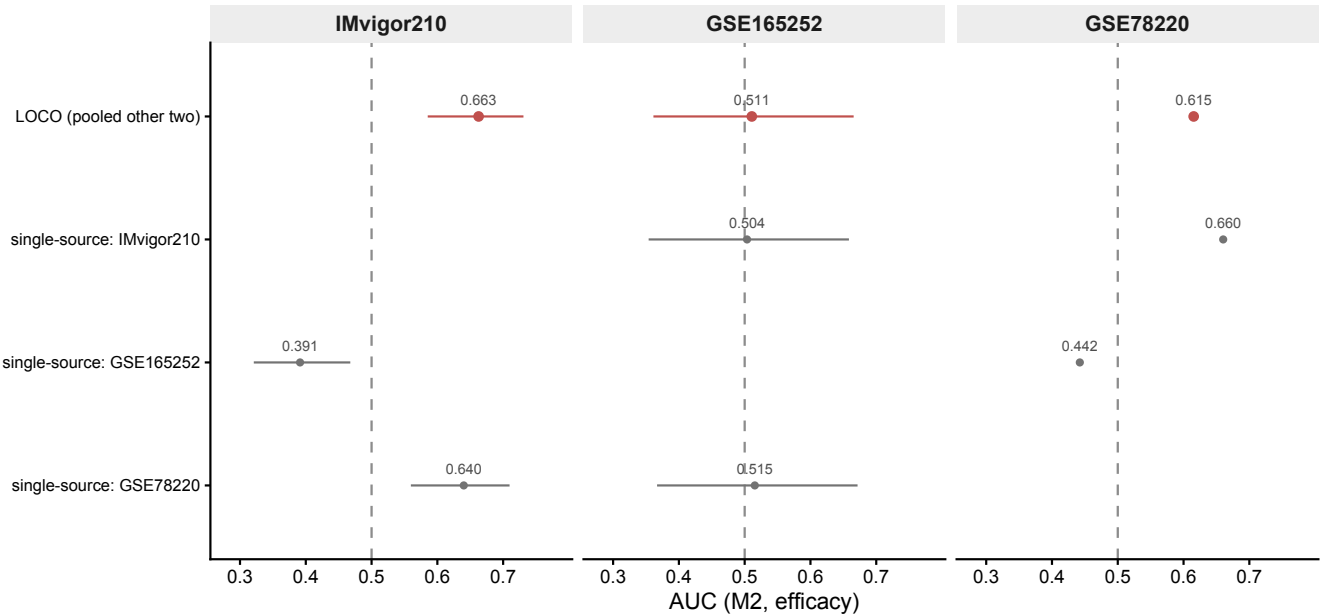
predictor RE-TRAINED on the pooled
other two cohorts for every target
→ a DIFFERENT prediction system
→ answers a different question
("does any 2-cohort mixture transfer?")

Why the distinction matters (v82 lock):

LOCO numbers CANNOT substitute for the primary claim: the LOCO M2 score for target IMvigor210 trains on GSE165252+GSE78220 — a predictor that never existed in the frozen pipeline. Primary-estimand transports (train IMvigor210, single-source, frozen) are the anchor-arm headline; LOCO M0–M6 (Fig. 5a) is a sensitivity benchmark over a wider strategy grid.

b

M2 under the two estimands, side by side (same target cohorts)



red = LOCO predictor (re-trained on pooled other two cohorts); grey = single-source frozen transports.
IMvigor210 panel: LOCO 0.663 and single-source GSE78220→IMvigor210 0.640 both exclude 0.5 — but these are DIFFERENT predictors from the primary anchor arm; neither licenses the frozen IMvigor210-anchored claim.